Question : 8 Date: 26/10/2026

Question 8: Monthly Loan Repayment Estimate

Phase 1 — COBOL Fundamentals | Difficulty: Beginner

Problem Statement

A bank employee wants a simple principal-only repayment estimate for a loan.

Task

Read the loan principal and repayment term in months. Calculate the equal monthly principal portion using division. Display two decimal places. This exercise excludes interest and fees.

Input Format

- Line 1: loan principal.
- Line 2: term in months.

Output Format

- Print the monthly principal repayment to two decimal places, or INVALID INPUT if the term is not positive.

Constraints

- 0 ≤ principal ≤ 100,000,000
- 1 ≤ months ≤ 600

Sample Input

```
120000
12
```

Sample Output

```
10000.00
```

Implementation Notes

Use decimal arithmetic and distinguish stored numeric precision from display formatting.

Algorithm:

Step 1: Start Program

Step 2: Defining inputs at Working storage under data division

Step 3: Getting inputs from user at main program under Procedure Division

Step 4: Verifying input constraints for month and amount

Step 5: Computing monthly repayment

Step 6: Moving values to temporary display value to type casting and remove leading zeros

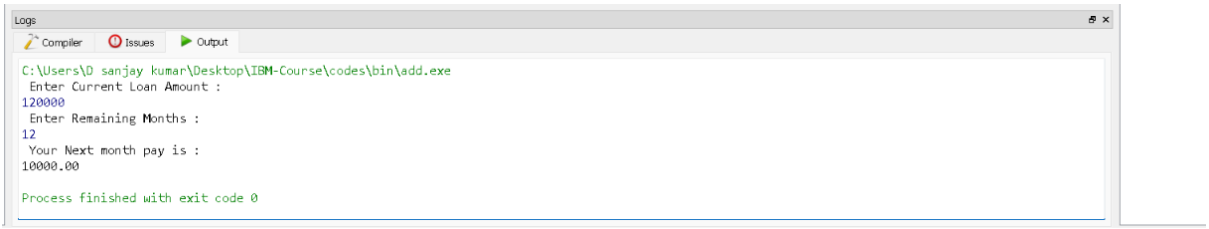
Step 7: Display Output

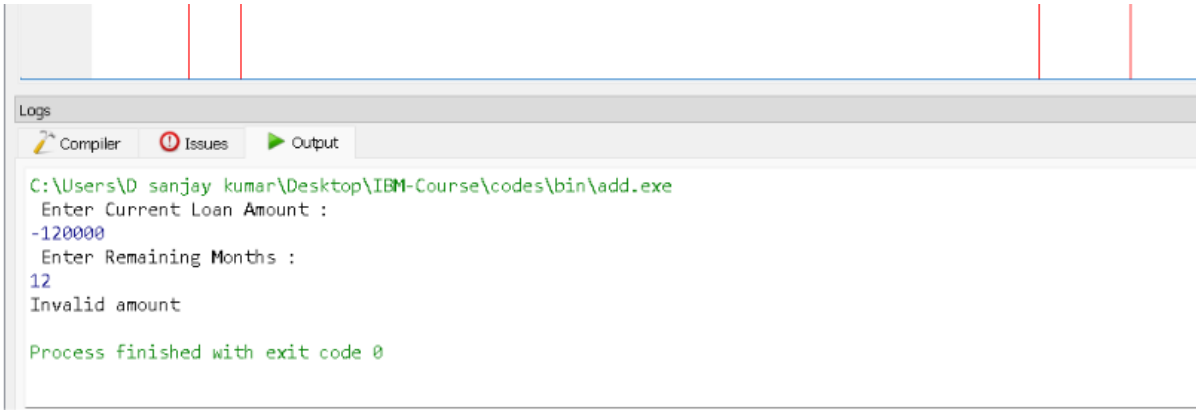
Step 8 : Exit program

Program:

```
IDENTIFICATION DIVISION.
PROGRAM-ID. Repayment-Estimate.
ENVIRONMENT DIVISION.
DATA DIVISION.
WORKING-STORAGE SECTION.
01 loan-amount PIC S9(9).
01 months PIC S9(2).
01 monthly-pay PIC 9(9)v99.
01 display-pay PIC zzzzzzz9.99.
PROCEDURE DIVISION.
MAIN-PROGRAM.
DISPLAY " Enter Current Loan Amount : "
ACCEPT loan-amount
DISPLAY " Enter Remaining Months : "
ACCEPT months
IF loan-amount > 0 AND loan-amount <= 100000000
IF months >= 0 AND months <= 600
COMPUTE monthly-pay = loan-amount / months
MOVE monthly-pay TO display-pay
DISPLAY " Your Next month pay is : "
DISPLAY FUNCTION TRIM(display-pay LEADING)
ELSE
DISPLAY " Invalid Month "
ELSE
DISPLAY "Invalid amount "
END-IF
STOP RUN.
```

Ouput:




Logs
Compiler
Issues
Output
C:\Users\D sanjay kumar\Desktop\IBM-Course\codes\bin\add.exe
Enter Current Loan Amount :
-120000
Enter Remaining Months :
12
Invalid amount
Process finished with exit code 0